Sir Syed C@SE Institute of Technology, Islamabad

Lab Final-Term Exam – Fall 2025

AI 2025 Programming for AI

Name:_____ **Reg. #:**_____

Total Marks: 50 **Time Allowed: 110 min.**

EXAM INSTRUCTIONS:

1- CHEATING IS NOT ALLOWED. Looking at someone else screen is also cheating.

2- Mobile phones, ChatGPT and internet usage is not allowed.

3- If you have any queries related to task, you can ask instructors only. Don't talk to each other until you are allowed, otherwise **5 marks will be immediately deducted.**

Question A: (10 marks)

Design a Python program that models a **Vehicle Management System** using classes and objects.

1. Create a base class named Vehicle with the following:
   - Private attributes: __brand and __price
   - A constructor to initialize these attributes
   - Getter and setter methods to access and update these private attributes (encapsulation)

2. Create a child class named Car that inherits from Vehicle and has:
   - An additional attribute fuel_type
   - A method display_details() that prints all vehicle details in a proper format

3. In the main program:
   - Create at least **two Car objects**
   - Update the price of one car using a setter method
   - Display details of both cars

Question B: (20 marks)

You are provided with a dataset named **car_evaluation_dataset.csv**.
Load the dataset using pandas and answer the following questions.

Q2(a) [2 marks]
Load the dataset and display:
- Ten random entries from a dataset
- total rows and columns of dataset in the form of a tuple

Q2(b) [2 marks]
Check for **missing values** in each column and display the result.

Q2(c) [3 marks]
Display the **count of each category** in the column class.

Q2(d) [3 marks]
Filter and display all records where:

- doors is "4"
- AND safety is "high"

Q2(e) [4 marks]
Perform simple encoding on the categorical column "doors" as follows:

Original Value	Encoded Value
"2"	2
"3"	3
"4"	4
"5more"	5

➔ Also Store the encoded values in a new column named doors_encoded.

Q2(f) [3 marks]

Convert the column doors_encoded into a **NumPy array** and:

- calculate the mean and median
- calculate the most occurring value

Q2(g) [3 marks]

Using NumPy, create a new array by multiplying each element of the doors_encoded array by 3.5.

Q2(h) [4 marks]

Create **one simple visualization**:

- a bar chart showing the count of cars for each category in the safety column and also in a "lug_boot" column.

Question C: (10 Marks)

Write a Python program to scrape data from the Cheezious website:

https://cheezious.com

1. Use the requests library to fetch the webpage.
2. Use BeautifulSoup to parse the HTML content.

3. Scrape and display:
    - restaurant or brand name
    - menu item names
    - prices
4. Store the scraped data in:
    - Csv file and save it

Note: 10 marks for viva.

***** Good Luck *****